

Basic Code Analysis: Identifying Memory Leaks & Resource Cleanup

Duration : 30 Min.

Scenario

A database connection pool exhausts available handles under load. Create 'db_pool.py' in student_workspace/ and add proper connection cleanup.

Target File Location & Creation

File to Create/Update: student_workspace/db_pool.py

Task Objectives

Perform the following actions inside the student_workspace directory: - Create a file named 'db_pool.py' inside 'student_workspace/'. Implement connection cleanup using 'try...finally: conn.close()' or a context manager.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect the scenario input files in student_workspace/.
3. Create/update the file student_workspace/db_pool.py and populate it with valid parameters and configurations.
4. Save your changes (Ctrl + S or Cmd + S).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your work by running python run.py in the terminal before submitting.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	db_pool.py exists in student_workspace/ with valid Python code	3 Marks

TC2	conn.close() or context manager added	4 Marks
TC3	Connection leak prevented on exceptions	3 Marks

Total Score: 10 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.